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Unit 5 : Session 3 : SLO 1 | Flynn's Classification

I. Multiple Choice

1. b) Number of instruction streams and data streams.
2. b) GPU executing the same instruction on multiple data points — classic SIMD.
3. b) Execute different instructions on different data.
4. a) Standard single-core processor.
5. d) MIMD — independent instruction and data streams scale to large clusters/multicores.

II. True or False

1. TRUE — SISD = single instruction, single data.
2. FALSE — MISD is rare/specialized (e.g. redundant flight controllers), not general-purpose.
3. TRUE — image/vector workloads apply one operation to many data elements.
4. TRUE — that is exactly MIMD.
5. TRUE — both are parallel models in Flynn's scheme.

III. Scenario-Based Activity

- System A — MIMD: multiple processors each running different instructions on different data (scientific simulation nodes).
- System B — SIMD: one rendering operation applied simultaneously to large pixel arrays (GPU-style).
- System C — SISD: a single-core embedded controller executing one instruction stream on one data stream.